

ball radius $r = 0.1m$

ball mass $m = 1kg$

Ball State: $q = \begin{bmatrix} r_0 \in \mathbb{R}^3 \\ Q_0 \in \mathbb{R}^4 \end{bmatrix}$ $V = \begin{bmatrix} \dot{r}_0 \in \mathbb{R}^3 \\ w_0 \in \mathbb{R}^3 \end{bmatrix}$ } linear velocity in inertial frame } angular velocity in body frame

kinetic energy: $T = \frac{1}{2}\|\dot{r}_0\|_2^2 + \frac{1}{2} w_0^T J w_0 = \frac{1}{2} \begin{bmatrix} \dot{r}_0 \\ w_0 \end{bmatrix}^T \underbrace{\begin{bmatrix} mI & \\ & J \end{bmatrix}}_{M(q)} \begin{bmatrix} \dot{r}_0 \\ w_0 \end{bmatrix}$

potential energy: $U = mg\, [0\ 0\ 1]\, r_0$

· Floating-Base Dynamics

$$\min_{r_0, Q_0, w_0} \int L(q,v)\, dt = \int \frac{1}{2} v^T M(q) v - U(q)\, dt$$

$$\text{s.t.} \quad \dot{Q}_0 = Q_0 * \begin{bmatrix} 0 \\ \frac{1}{2} w_0 \end{bmatrix} \qquad w_0 = 2H^T L(Q_0)^T \dot{Q}_0$$

$$\partial / \partial \begin{bmatrix} r_0 \\ Q_0 \\ w_0 \end{bmatrix} \int \frac{1}{2} v^T M(q) v - U(q) + l^T \left(2H^T L(Q_0)^T \dot{Q}_0 - w_0 \right) dt$$

1° $\frac{\partial}{\partial r_0} \int L(q,v)\, dt = 0$

$\frac{d}{dt} \frac{\partial L}{\partial \dot{r}_0} - \frac{\partial L}{\partial r_0} = 0$ Euler-Lagrange Equation

2° $\frac{\partial}{\partial w_0} \int L(q,v) - l^T w_0\, dt = 0$

$l = \frac{\partial L}{\partial w_0}$

3° $\frac{\partial}{\partial Q_0} \int L(q,v) + 2 l^T H^T L(Q_0)^T \dot{Q}_0\, dt = 0$

$$0 = \frac{\partial}{\partial \epsilon} \int L\left(\begin{bmatrix} r_0 \\ Q_0 + \epsilon \delta Q_0 \\ w_0 \end{bmatrix}, v \right) + 2 l^T H^T L(Q_0 + \epsilon \delta Q_0)^T (\dot{Q}_0 + \epsilon \dot{\delta Q}_0)\, dt \Big|_{\epsilon = 0}$$

$$= \int \frac{\partial L}{\partial Q_0}^T \delta Q_0\, dt + \int l^T [w] \delta\phi\, dt - \int \dot{l}^T \delta\phi\, dt$$

$$= \int \frac{\partial L}{\partial Q_0}^T \frac{1}{2} L(Q_0) H \delta\phi\, dt + \int l^T [w] \delta\phi\, dt - \int \dot{l}^T \delta\phi\, dt$$

$\frac{1}{2} H^T L(Q_0)^T \frac{\partial L}{\partial Q_0} - [w] l - \dot{l} = 0$

$\frac{d}{dt} \frac{\partial L}{\partial w_0} + w \times \frac{\partial L}{\partial w_0} - \underbrace{\frac{1}{2} H^T L(Q_0)^T \frac{\partial L}{\partial Q_0}}_{\frac{\partial L}{\partial Q_0} = 0 \text{ in this example}} = 0$

4° combine

$\frac{d}{dt} \boxed{\frac{\partial L}{\partial \dot{r}_0}} - \frac{\partial L}{\partial r_0} = 0$

$\frac{d}{dt} \frac{\partial L}{\partial w_0} + w_0 \times \frac{\partial L}{\partial w_0} = 0$

$\frac{\partial L}{\partial v} = M(q) v$

$\frac{d}{dt} \frac{\partial L}{\partial v} = M(q)\dot{v} + \underbrace{\frac{\partial}{\partial q}(M(q)v)\dot{q}}_{\frac{\partial}{\partial q}(M(q)v) = 0 \text{ in this example}}$

$$M(q)\dot{v} + \begin{bmatrix} -\frac{dL}{dr_0} \\ w_0 \times \frac{dL}{dw_0} \end{bmatrix} = \frac{dq}{ds}^T \underbrace{Dr^{ij}(q)^T} F_{i,ext}$$

$r^{ij}(q)$ is the contact point

$F_{i,ext}$ is contact force + friction force

work done is $F_{i,ext}^T Dr^{ij}(q)\, \delta q = F_{i,ext}^T Dr^{ij}(q) \frac{dq}{ds} \underbrace{\delta s}$ virtual displacement in V space

work done by $F_{i,ext}$ is $F_{i,ext}^T \delta r^{ij} = F_{i,ext}^T \begin{bmatrix} \delta r_0 + (\delta\phi \times (r^{ij} - r_0)_{/0})_{/i} \end{bmatrix}$

1° translation work: $\frac{\partial}{\partial \delta r_0} \left(F_{i,ext}^T \delta r_0 \right) = F_{i,ext}$

2° rotation work: $\frac{\partial}{\partial \delta\phi} \left[(\delta\phi \times (r^{ij} - r_0)_{/0})_{/i}^T F_{i,ext} \right] = \frac{\partial}{\partial \delta\phi} \left[(\delta\phi \times (r^{ij} - r_0)_{/0})^T F_{i,ext/0} \right]$

$= \frac{\partial}{\partial \delta\phi} \left[(-[(r^{ij} - r_0)_{/0}]\, \delta\phi)^T F_{i,ext/0} \right]$

$= -[(r^{ij} - r_0)_{/0}]^T F_{i,ext/0}$

$= [(r^{ij} - r_0)_{/0}]\, H^T L(\alpha_0)^T R(\alpha_0)\, H\, F_{i,ext}$

$$M(q)\dot{v} + \underbrace{\begin{bmatrix} -\frac{dL}{dr_0} \\ w_0 \times \frac{dL}{dw_0} \end{bmatrix}}_{C(q,v)} = \underbrace{\begin{bmatrix} I \\ [(r^{ij} - r_0)_{/0}]\, H^T L(\alpha_0) R(\alpha_0)^T H \end{bmatrix}}_{(Dr^{ij}(q) \frac{dq}{ds})^T} F_{i,ext}$$

- Constraints

contact point in body frame

the contact point is $r^{ij}(q) = \begin{cases} r_0 + Q_0 * \underbrace{\begin{bmatrix} 0 \\ 0 \\ d \end{bmatrix}} * Q_0^{-1} \\ r_0 + \begin{bmatrix} 0 \\ 0 \\ -r \end{bmatrix} \end{cases}$

the contact constraint is $\phi^{ij}(r^{ij}(q)) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}^T r^{ij}(q) \geq 0$

the normal space at r^{ij} is $n^{ij} = \nabla\phi^{ij}(r^{ij}(q)) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

the tangent space at r^{ij} is $T^{ij} = \begin{bmatrix} t_1^{ij} & t_2^{ij} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$

the tangent velocity at r^{ij} is $V_t^{ij} = T^{ij\,T} Dr^{ij}(q)\, \dot{q} = T^{ij\,T} Dr^{ij}(q) \frac{dq}{ds} v$

the friction force at r^{ij} is $T^{ij} \underbrace{\lambda_t^{ij}}$ friction in T^{ij} coordinate

the contact force at r^{ij} is $n^{ij} \lambda_n^{ij}$

Lagrange Dynamics

$$M(q)\dot{v} + C(q,v) = \frac{dq}{ds}^T Dr^{ij}(q)^T \begin{bmatrix} n^{ij} & T^{ij} \end{bmatrix} \begin{bmatrix} \lambda_n^{ij} \\ \lambda_t^{ij} \end{bmatrix}$$

$$= \underbrace{J_n^{ij\,T}} \lambda_n^{ij} + J_t^{ij\,T} \lambda_t^{ij+} - J_t^{ij\,T} \lambda_t^{ij-}$$

normal jacobian: $J_n^{ij} = n^{ij\,T} Dr^{ij}(q) \frac{dq}{ds}$ $\quad J_t^{ij} = T^{ij\,T} Dr^{ij}(q) \frac{dq}{ds}$

Contact Complementarity $\quad 0 \leq \phi^{ij}(r^{ij}) \perp \lambda_n^{ij} \geq 0$

Friction complementarity $\quad 0 \leq \begin{bmatrix} \lambda_n^{ij} \\ \lambda_t^{ij+} \\ \lambda_t^{ij-} \end{bmatrix} \perp \begin{bmatrix} V_t^{ij} + f^{ij} 1 \\ -V_t^{ij} + f^{ij} 1 \\ \mu^{ij} \lambda_n^{ij} - 1^T (\lambda_t^{ij+} + \lambda_t^{ij-}) \end{bmatrix} \geq 0$

① Discrete Lagrange Dynamics (forward in time)

$$M(q_k)\left(\frac{V_{k+1}-V_k}{h}\right) + C(q_k,V_k) = J_n^{\alpha_j\top}\lambda_n^{\alpha_j} + J_t^{\alpha_j\top}\lambda_t^{\alpha_j+} - J_t^{\alpha_j\top}\lambda_t^{\alpha_j-}$$

$$V_{k+1} = V_k + h\,M(q_k)^{-1}\left[J_n^{\alpha_j\top}\lambda_n^{\alpha_j} + J_t^{\alpha_j\top}\lambda_t^{\alpha_j+} - J_t^{\alpha_j\top}\lambda_t^{\alpha_j-} - C(q_k,V_k)\right]$$

② Discretize q (backward in time)

$$q_{k+1} = q_k + h\frac{\partial q}{\partial s}V_{k+1}$$

③ Linearize ϕ (backward in time)

$$\phi^{\alpha_j}(r^{\alpha_j}(q_{k+1})) \approx \phi^{\alpha_j}(r^{\alpha_j}(q_k)) + h\cdot \underbrace{\frac{\partial\phi^{\alpha_j}}{\partial r^{\alpha_j}}(r^{\alpha_j}(q_{k+1}))\,Dr^{\alpha_j}(q_k)\frac{\partial q}{\partial s}}_{J_n^{\alpha_j}} V_{k+1}$$

$$= \phi^{\alpha_j}(r^{\alpha_j}(q_k)) + h\,J_n^{\alpha_j}\left\{V_k + h\,M(q_k)^{-1}\left[J_n^{\alpha_j\top}\lambda_n^{\alpha_j} + J_t^{\alpha_j\top}\lambda_t^{\alpha_j+} - J_t^{\alpha_j\top}\lambda_t^{\alpha_j-} - C(q_k,V_k)\right]\right\}$$

$$= \phi^{\alpha_j}(r^{\alpha_j}(q_k)) + hJ_n^{\alpha_j}V_k + h^2 J_n^{\alpha_j}M(q_k)^{-1}\left[J_n^{\alpha_j\top}\lambda_n^{\alpha_j} + J_t^{\alpha_j}\lambda_t^{\alpha_j+} - J_t^{\alpha_j\top}\lambda_n^{\alpha_j} - C(q_k,V_k)\right]$$

$$\frac{1}{h}\phi^{\alpha_j}(r^{\alpha_j}(q_{k+1})) = \frac{1}{h}\phi^{\alpha_j}(r^{\alpha_j}(q_k)) + J_n^{\alpha_j}V_k - hJ_n^{\alpha_j}M(q_k)^{-1}C(q_k,V_k) + h\left[\underbrace{J_{nn}^{\alpha_j}}\lambda_n^{\alpha_j} + J_{nt}^{\alpha_j}\lambda_t^{\alpha_j+} - \underbrace{J_{nt}^{\alpha_j}}\lambda_t^{\alpha_j-}\right]$$

$$J_{nn}^{\alpha_j} = J_n^{\alpha_j}M(q_k)J_n^{\alpha_j\top} \qquad J_{nt}^{\alpha_j} = J_n^{\alpha_j}M(q_k)^{-1}J_t^{\alpha_j\top}$$

④ tagent velocity (backward in time)

$$V_t^{\alpha_j} = T^{\alpha_j\top}Dr^{\alpha_j}(q_k)\,q_{k+1}$$

$$= T^{\alpha_j\top}Dr^{\alpha_j}(q_k)\frac{\partial q}{\partial s}V_{k+1}$$

$$= J_t^{\alpha_j}V_{k+1}$$

$$= J_t^{\alpha_j}\left\{V_k + h\,M(q_k)^{-1}\left[J_n^{\alpha_j\top}\lambda_n^{\alpha_j} + J_t^{\alpha_j\top}\lambda_t^{\alpha_j+} - J_t^{\alpha_j\top}\lambda_t^{\alpha_j-} - C(q_k,V_k)\right]\right\}$$

$$= J_t^{\alpha_j}V_k - hJ_t^{\alpha_j}M(q_k)^{-1}C(q_k,V_k) + h\left[J_{tn}^{\alpha_j}\lambda_n^{\alpha_j} + J_{tt}^{\alpha_j}\lambda_t^{\alpha_j+} - J_{tt}^{\alpha_j}\lambda_t^{\alpha_j-}\right]$$

$$0 \le \begin{bmatrix}\lambda_n^{\alpha_j}\\ \lambda_t^{\alpha_j+}\\ \lambda_t^{\alpha_j-}\\ \gamma^{\alpha_j}\end{bmatrix} \perp \begin{bmatrix}\frac{1}{h}\phi^{\alpha_j}(r^{\alpha_j}(q_{k+1}))\\ V_t^{\alpha_j} + \gamma^{\alpha_j}\mathbf{1}\\ -V_t^{\alpha_j} + \gamma^{\alpha_j}\mathbf{1}\\ \mu^{\alpha_j}\lambda_n^{\alpha_j} - \mathbf{1}^\top(\lambda_t^{\alpha_j+} + \lambda_t^{\alpha_j-})\end{bmatrix} \ge 0$$

$$0 \le \begin{bmatrix}\lambda_n^{\alpha_j}\\ \lambda_t^{\alpha_j+}\\ \lambda_t^{\alpha_j-}\\ \gamma^{\alpha_j}\end{bmatrix} \perp \begin{bmatrix} hJ_{nn}^{\alpha_j} & hJ_{nt}^{\alpha_j} & -hJ_{nt}^{\alpha_j} & 0\\ hJ_{tn}^{\alpha_j} & hJ_{tt}^{\alpha_j} & -hJ_{nt}^{\alpha_j} & \mathbf{1}\\ -hJ_{tn}^{\alpha_j} & -hJ_{tt}^{\alpha_j} & hJ_{tt}^{\alpha_j} & \mathbf{1}\\ \mu^{\alpha_j} & -\mathbf{1}^\top & -\mathbf{1}^\top & 0\end{bmatrix}\begin{bmatrix}\lambda_n^{\alpha_j}\\ \lambda_t^{\alpha_j+}\\ \lambda_t^{\alpha_j-}\\ \gamma^{\alpha_j}\end{bmatrix} + \begin{bmatrix}\frac{1}{h}\phi^{\alpha_j}(r^{\alpha_j}(q_k)) + J_n^{\alpha_j}V_k - hJ_n^{\alpha_j}M(q_k)^{-1}C(q_k,V_k)\\ J_t^{\alpha_j}V_k - hJ_t^{\alpha_j}M(q_k)^{-1}C(q_k,V_k)\\ -J_t^{\alpha_j}V_k + hJ_t^{\alpha_j}M(q_k)^{-1}C(q_k,V_k)\\ 0\end{bmatrix} \ge 0$$